Integrating Education Research into Atmospheric Science Field Campaigns

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Educational Experiences in Atmospheric Science Field Campaigns A Community Workshop

Colorado State University, May 2025



Figure 1. Workshop attendees

Workshop Motivation

Atmospheric Science Field Campaigns provide unique experiential observational and learning opportunities for students as they occur during short windows time, require a mixture of expertise, can be a pivotal career moment and offer unique network and mentoring opportunities. Led by the members of the Atmospheric Science Education Research (ASER) community, this workshop brought experts from three domains together (Fig. 2) to discuss:

- How do we ensure that student learning occurs?
- How do we maximize success, as participating students depend on data collected during the campaign for graduation?
- How do we ensure inclusion and access for a broad range of learners in field campaigns?

Workshop Objectives

1. Integrate field campaign research and opportunities with an investigation of student learning.
2. Define common goals for advancing our understanding of how students and early career scientists think and learn through authentic, experiential learning in the discipline.

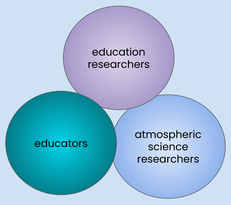


Figure 2: Expertise of workshop attendees

Workshop Day 1
Field Trip to NSF NCAR Research Aviation Facility
IGNITE Talks: ASER Scholarly Activities and Campaign Research Activities
Gallery Walk: Deconstructing A Field Campaign



Figure 3: Workshop attendees visiting the NSF NCAR Research Aviation Facility to gain insights into the logistics of large airborne measurement campaigns.



IGNITE talks: ASER Scholarly Activities and Campaign Research Activities



Figure 4: Day 1 IGNITE talks

Dawn Kopacz: Background and examples of ASER research
Emily Ward: Student Field Experiences: Assessing Outcomes of student learning
Kathy Quardokus Fisher: Building transferability and/or generalizability in ASER

Zeljka Stone: OTREC Airborne campaign
Shane Murphy: Ground based campaigns
Steve Nesbitt: RELAMPAGO and International campaigns

Gallery Exploration: Deconstructing the Field Campaign

Prompts:

- What could students and trainees learn during each phase?
- What are unique education research questions that could arise during each phase?



Figure 5: Attendees participating in the gallery walk



Overall Concept



Scientific proposal & Identifying science questions



Post selection planning



Deploying the field campaign



Post- campaign science team activities

Figure 6: Collective input from the gallery walk

Workshop Day 2
Working Groups by key themes identified in the De-constructing session
World Cafe: Visiting other working groups
Group Presentations

Key Themes from "De-constructing Session"

1. Ideation, Research Design, Proposal Development, & Writing
2. Team Building & Collaboration
3. Skills Development, Content & Assessment
4. Cultural Integration & Sense of Belonging
5. Long term outcomes: Professional Identity and Broader Impacts
6. Maximize Participation & Increase Accessibility
7. Creativity, Adaptability, and Resiliency





Figure 7: Working groups exploring their themes

Working Group Session:

Define specific, actionable research questions related to integrating education and research in field campaigns.

Determine what evidence would demonstrate the effectiveness of different educational approaches in field campaigns.

Discuss what have you experienced or noticed when including trainees in the field? Classroom learning to real experience?

Develop a detailed action plan outlining how to implement the research and integrate education into future field campaigns.

Session Outcome: Working groups develop a concept map of their ideas

Sustaining a Community of Practice: Communication, Connection, and Continuity

If you are interested in joining this new community, please contact us. And scan QR code to learn more and to read the full Meeting Summary



Workshop Day 3 (half day)
Working Group Takeaways: Community Statement
Report Outs
Reflections

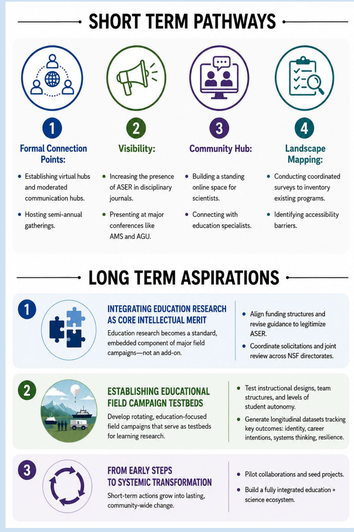
Community Statement

1. A Vision for Integrating Physical Science and Education Research in Field Settings
2. Short-Term Pathways and Long-Term Aspirations
3. Strengthening Integration Between Education Researchers and Field Campaign Scientists
4. Sustaining a Community of Practice: Communication, Connection, and Continuity

Scan QR code to view full Community Statement

A Vision for Integrating Physical Science and Education Research in Field Settings

Position ASER as a collaborative research endeavor during observational field campaigns.	Ensure opportunities for **co-development and co-production of research goals** between physical and education researchers.	Contribute to **inclusivity, skill building, content knowledge**, and engagement through equitable and accessible field experiences.
Foster interdisciplinary fluency and leadership, particularly among early-career researchers.	Develop an **open-access ASER Toolbox** that houses resources for designing, implementing, and publishing education research across field campaigns.	Establish **sustained mechanisms for collaboration**, reflection, and improvement.



Key Readings and Resources

A Road Map to Success of International Field Campaigns in Atmospheric and Oceanic Sciences
Zhang and Moore (2023)

Enlightenment Strikes! Broadening Graduate School Training through Field Campaign Participation
Rasmussen et al. (2021)

How Undergraduate Students Learn Atmospheric Science: Characterizing the Current Body of Research
McNeal et al. (2022)

NSF NCAR Supported Field Projects: Videos from EdEC

(each site has videos about the campaign, as well as education related videos)

RELÁMPAGO: Remote Sensing of Electrification, Lightning, and Mesoscale/Microscale Processes with Adaptive Ground Observations

OTREC: Organization of Tropical East Pacific Convection

CAESAR: Cold-Air Outbreak Experiment In the Sub-Arctic Region

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Image Credit: All photos used throughout were taken by M. Burt.